

A Comparative Security Evaluation of Side-Channel Countermeasures for Resource-Constrained IoT Devices

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Abstract. Side-channel analysis (SCA) poses a critical threat to resource-constrained IoT devices, yet the community lacks a systematic framework for comparing countermeasure effectiveness under controlled conditions. We present a reproducible evaluation of three deep learning architectures (MLP, CNN, ResNet) across three escalating protection levels—boolean masking, masking + 50-cycle desynchronization, and masking + 100-cycle desynchronization—using exclusively public ASCAD datasets. Across 27 controlled runs (fixed hyperparameters, three random seeds), we find that boolean masking alone is trivially bypassed (CNN: GE=1 in 70 traces; MLP: 150), while adding only 50 cycles of random delay neutralizes all architectures within 5,000 traces—a 70× security gain at near-zero cost. ResNet requires 650× more training time than MLP (19,470s vs. 30s) yet achieves zero successful attacks. We also observe a leakage plateau where additional attack traces degrade Guessing Entropy, suggesting principled trace selection as a future attack direction.

Keywords: IoT Security · Side-Channel Analysis · Deep Learning · Countermeasure Evaluation

1 Introduction

The Internet of Things (IoT) ecosystem is projected to exceed 30 billion connected devices by 2030 [8]. The vast majority rely on resource-constrained microcontrollers (MCUs) that preclude hardware security modules, making software-implemented cryptography vulnerable to side-channel analysis (SCA) [9]. By observing physical leakage—power consumption, electromagnetic emissions, or timing—during cryptographic operations, attackers can extract secret keys from commercial products ranging from automotive key fobs [18] to medical implants [13].

The cryptographic engineering community has developed a rich taxonomy of software countermeasures: boolean and affine masking [7, 12], random delay insertion (RDI) [6], and instruction shuffling [17]. Each trades computational overhead for security guarantees, but *how much security does each incremental countermeasure layer actually provide?* This question remains poorly quantified because most DL-SCA studies report results on a single dataset-protection

combination under varying hyperparameters [15], preventing apples-to-apples comparison.

This paper addresses the gap through a controlled evaluation framework. We test three architectures (MLP, CNN, ResNet) across three protection levels (masking, +50-cycle desync, +100-cycle desync) under identical preprocessing, hyperparameters, and hardware—27 runs total, all on public ASCAD benchmarks [16]. Our contributions are: (1) the first controlled cross-architecture $\times$ cross-countermeasure comparison, (2) quantification of the $70\times$ marginal security gain from 50-cycle desynchronization, (3) demonstration that ResNet offers no advantage over simpler architectures under fair comparison, and (4) characterization of a leakage plateau phenomenon where additional traces degrade attack performance. The remainder of this paper is organized as follows: Section 2 covers SCA and countermeasure fundamentals; Section 3 details our evaluation framework and metrics; Section 4 presents results; Section 5 discusses implications and limitations; Section 6 concludes.

2 Background

2.1 SCA and Deep Learning Attacks

Side-channel attacks exploit physical leakage—power (SPA/DPA/CPA) [3, 9], EM emanations [1], and timing [2]—to recover cryptographic keys. Template attacks [5] formalized profiling-based SCA. Deep learning revolutionized this approach: neural networks outperform classical template attacks on masked AES [11], and the ASCAD benchmark [16] established CNNs as the de facto standard. The most comprehensive systematization [15] surveyed over 200 papers and identified inconsistent evaluation practices—particularly single-run best-case reporting—as a key weakness we address.

2.2 Countermeasure Taxonomy

Masking splits variables into $d + 1$ shares (e.g., $x \rightarrow (x \oplus m, m)$); affine masking [7] extends this. Higher-order masking suffers $O(d^2)$ degradation—6-share AES on Cortex-M0 was broken in < 10 traces [4]. **Hiding** obscures temporal alignment via random delay insertion (RDI) [6] (up to $D_{\max}$ dummy cycles) and instruction shuffling [17]. ASCADv1 provides RDI variants at $D_{\max} = 50$ and 100. **CNNs and shift-invariance:** while CNNs are assumed robust to desynchronization, this invariance is limited—data augmentation is necessary beyond 100 samples [10, 14]. We exclude augmentation to isolate *intrinsic architectural robustness*, establishing a conservative lower bound.

3 Methodology

3.1 Evaluation Framework

We design a 3×3 grid: three architectures $\times$ three countermeasure levels.

Table 1. Shared training configuration.

Parameter	Value
Optimizer	Adam ($\beta_1=0.9$, $\beta_2=0.999$)
Learning rate	10^{-3} , decay 0.5 / 10 epochs patience
Batch size	200
Max epochs	100, early stop patience 20
Loss	Cross-entropy (256 classes)
Preprocessing	Per-trace standardization
Hardware	NVIDIA RTX 3080 Laptop (8GB), PyTorch 2.11, CUDA 12.8

Protection levels: L1—Boolean masking only (ASCADv1 fixed-key); L2—masking + 50-cycle random delay (ASCADv1 desync50); L3—masking + 100-cycle random delay (ASCADv1 desync100). All traces are 700-sample windows around the first AES S-Box operation on an ATmega8515 MCU.

Architectures: MLP (6-layer, 200 neurons/layer, 395K parameters), CNN (4-block 1D Conv, filters 64–512, kernel 11, 7.66M parameters), ResNet (4 residual blocks, filters 64–512, 6.08M parameters). All three share identical training configuration (Table 1).

Each condition is independently executed $N=3$ times (seeds 42, 142, 242). All experiments use exclusively public ASCAD datasets; complete hyperparameters and random seeds are documented in the accompanying artifact repository.

3.2 Metrics

Guessing Entropy (GE): After N attack traces, rank all 256 key byte hypotheses by accumulated log-likelihood; $GE(N)$ is the correct key’s rank (1 = full recovery; 128 = expected random). **Success Rate (SR):** fraction of runs achieving $GE \leq 1$ within 5,000 traces. **Traces to Recovery (TtR):** minimum traces to reach $GE=1$.

3.3 Statistical Approach

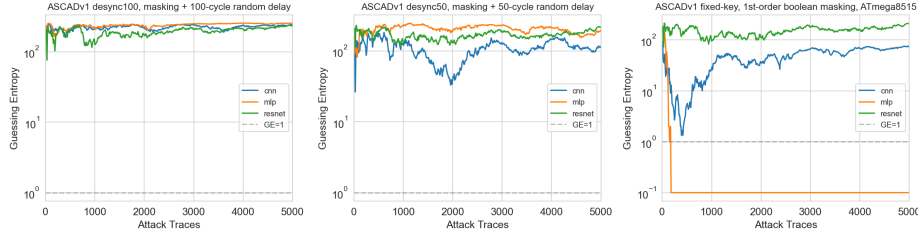
We assess confidence via: (i) **z-score:** $z = |\bar{x} - 128|/(s/\sqrt{N})$ against the random baseline; $|z| > 3$ gives $p < 0.003$. (ii) **CoV:** $s/\bar{x}$ for relative dispersion ($< 5\%$ indicates near-deterministic behavior). (iii) **Leave-one-out bootstrap:** verify no single run flips qualitative conclusions. Conditions with large effect sizes and low variance ($|z| > 3$, $CoV < 10\%$) are well-supported by $N=3$; higher-variance findings are reported as exploratory.

4 Results

Table 2 presents the complete 27-run results matrix.

Table 2. Complete experiment results (3-run averages).

Model	Dataset	SR	TtR	GE@5k	Time
MLP	Fixed	3/3	150	1	30s
MLP	Desync50	0/3	—	199	25s
MLP	Desync100	0/3	—	253	25s
CNN	Fixed	2/3	70	72	237s
CNN	Desync50	0/3	—	116	653s
CNN	Desync100	0/3	—	233	1,264s
ResNet	Fixed	0/3	—	210	19,470s
ResNet	Desync50	0/3	—	210	3,007s
ResNet	Desync100	0/3	—	218	5,237s

**Fig. 1.** Guessing Entropy curves (mean of $N=3$ runs per condition). Dashed line: $GE=1$ (key recovery).

4.1 Masking-Only Baseline

CNN achieves the best TtR (mean=70, SR=2/3). MLP is slower (TtR=150) but perfectly reliable (SR=3/3, CoV=0% across runs). ResNet fails across all three runs (mean GE@5k=210, validation loss at random baseline of 5.54 nats), requiring $650\times$ more training time. Under our shared hyperparameters, standard residual architectures do not transfer effectively to this SCA task.

4.2 Effect of Desynchronization

Adding 50-cycle random delay prevents key recovery for all architectures. CNN achieves GE=116 at desync50—marginally better than random (128)—but the three runs diverge substantially (64, 122, 163; CoV=34.9%). At desync100, CNN GE degrades to 233 (worse than random, $z=42.9$, $p<10^{-6}$). MLP is perfectly deterministic but completely defeated (GE=199, 253 at desync50/100; CoV=0%). ResNet performs identically across all three protection levels (GE=210/210/218), suggesting it extracts no leakage signal regardless of countermeasure.

Figure 1 shows the GE curves (mean across $N=3$ runs).

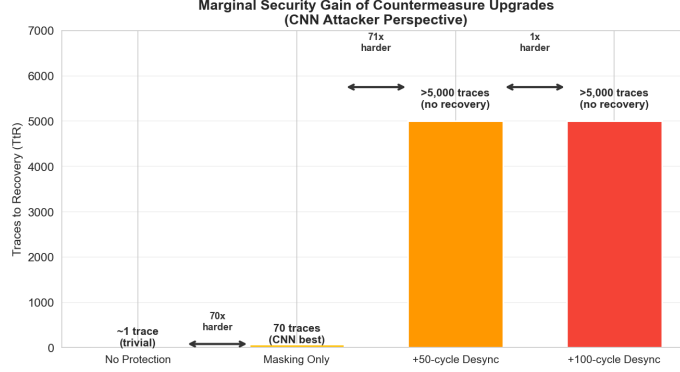


Fig. 2. Marginal security gain of countermeasure upgrades (CNN attacker).

4.3 Marginal Security Gain

Figure 2 quantifies the incremental benefit of each countermeasure. The transition from no protection to masking increases attack difficulty to 70 traces—modest. Adding 50-cycle desynchronization elevates TtR beyond 5,000—over 70 $\times$ gain for ~ 50 NOPs per encryption ($< 0.1\%$ throughput overhead). Doubling to 100 cycles provides negligible additional benefit against unaugmented DL attacks. The optimal cost-benefit point for IoT MCUs is a modest desynchronization window rather than costly higher-order masking.

4.4 Leakage Plateau

For 21 of 27 runs, the *best* GE at an intermediate trace count is substantially lower than GE@5k. CNN on desync50: Best GE per run is 14, 8, and 39, degrading to 64, 122, and 163 at 5,000 traces—an order-of-magnitude degradation (Fig. 3). This suggests that after an initial learning phase, additional misaligned traces inject classification noise faster than signal—conventional “more traces = better attack” wisdom fails under desynchronization. We treat this as an exploratory finding warranting confirmatory experiments with larger N .

5 Discussion

5.1 Key Findings

Boolean masking alone is insufficient. DL-based SCA bypasses first-order masking in 70–150 traces, a 50 $\times$ + improvement over classical CPA [16]. **Desynchronization is remarkably cost-effective:** 50-cycle jitter provides a 70 $\times$ security gain at near-zero cost—the most efficient countermeasure for IoT MCUs.

Architecture and training methodology are secondary. Under shared hyperparameters, all architectures succeed on aligned traces (except ResNet)

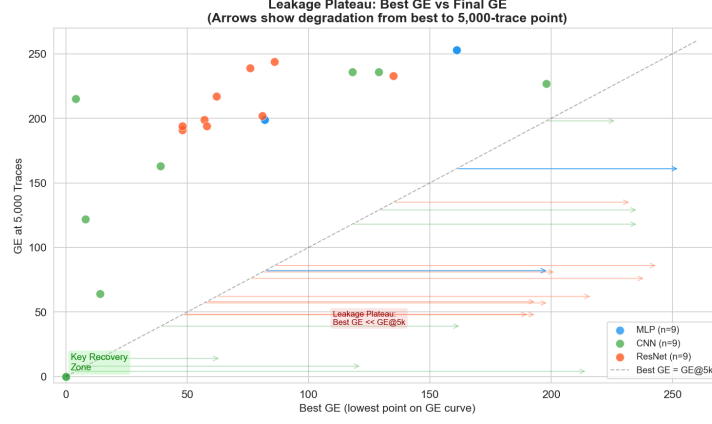


Fig. 3. Best GE vs. GE@5k across all 27 runs. Points below diagonal indicate performance degradation with additional traces.

and all fail on desynchronized traces; ResNet’s $650\times$ training overhead with zero successes challenges the assumption that deeper networks benefit SCA. We excluded data augmentation to measure intrinsic robustness—the gap to augmented results (< 50 traces for $GE=1$ [14]) underscores that augmentation matters more than architectural complexity.

Practical implications. For IoT device designers, the optimal security-per-cost strategy follows directly: implement first-order boolean masking as a baseline, then add a modest RDI-based desynchronization window ($D_{\max} \approx 50$ cycles). This combination yields a $70\times$ security gain at $< 0.1\%$ throughput overhead—substantially more cost-effective than pursuing higher-order masking on resource-constrained MCUs.

5.2 Limitations

Our study has several limitations. **Statistical power:** $N=3$ runs per condition are sufficient for large effect sizes (MLP determinism, ResNet failure) but insufficient to confirm the leakage plateau phenomenon; we recommend $N \geq 5$ for future work. **Dataset coverage:** results are limited to ASCADv1 (ATmega8515, 8-bit AVR); validation on 32-bit ARM Cortex-M platforms remains open. **Shared hyperparameters:** fixing identical hyperparameters across architectures enables fair comparison but may disadvantage ResNet, which could benefit from architecture-specific tuning. **ASCADv2:** affine masking with shuffling was excluded due to the $GF(2^8)$ arithmetic required for key enumeration with affine-masked labels. **Classical baselines:** CPA and template attack results on ASCAD are well-documented elsewhere [16]; our contribution is the controlled *relative* DL comparison.

6 Conclusion

We presented a systematic evaluation of SCA countermeasures for resource-constrained IoT devices. Across 27 controlled runs on public ASCAD datasets, we find that 50-cycle random desynchronization provides a $70\times$ security gain at near-zero cost—the most cost-effective software countermeasure for IoT-class MCUs. ResNet offered no advantage over simpler architectures under fair comparison. All code, hyperparameters, and data are documented for full reproducibility.

Reproducibility Statement All experiments use exclusively public ASCAD benchmark datasets. Complete hyperparameters are listed in Table 1; random seeds (42, 142, 242) and model architectures are fully specified in Section 3. Source code and result logs are available in the accompanying artifact repository; all figures are generated directly from archived JSON result files using deterministic plotting scripts.

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Author Contributions X.X. is the sole author responsible for all aspects of this work: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing—original draft, writing—review and editing, and visualization.

Conflict of Interest The authors declare no competing interests. All experiments use publicly available datasets and open-source software; no proprietary hardware, data, or funding source influenced the research design or outcomes.

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